

# A Guide to the Algorithmic Redistricting Portfolio

Giovanni Della Libera

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# 1 Overview

Congressional redistricting in the United States suffers a legitimacy crisis: partisan mapmakers manipulate district boundaries to predetermine election outcomes, eroding public confidence in representative democracy. This portfolio demonstrates that **recursive graph bisection**—a computational method used for supercomputer workload distribution—provides an algorithmic alternative that *cannot be instructed to gerrymander because it cannot access partisan data*.

Applied to all 50 states across three census decades (2000/2010/2020), the algorithm produces 435 congressional districts with near-perfect population equality, geographic contiguity, and geometric compactness. The research program comprises **190+ papers across twenty active tracks (A–G, I–U)**, spanning algorithm foundations (Track B), validation (Track C), legal compliance (Track D), experimental extensions (Tracks E–F), ensemble-based plan evaluation (Track G), incumbency and apportionment (Tracks I–J), compactness and partisan fairness metrics (Tracks K–L), community character weights (Track M), plan construction (Track T), and search and optimization (Track U), with Track A providing cross-portfolio synthesis.

Papers in this program are at varying stages of completion. Track B foundations, several Track G ensemble papers, the former Track H papers now merged into Track U, and selected Track T and U papers are accepted or ready for submission. Remaining papers across all tracks are in draft. The overall research program is ongoing.

## Four Core Contributions

- **Philosophical:** *Impossibility defense*—the algorithm cannot see partisan data, therefore cannot be instructed to gerrymander by construction. Compact neutral maps can still reflect geographic voter concentration, as shown in Track C.5.
- **Technical:** Edge-weighted graph partitioning achieves +22–44% compactness improvement over enacted maps, with seed variance below 2% for 96% of states.
- **Empirical:** National-scale validation across 50 states  $\times$  3 census decades (150 redistricting scenarios), with adversarial robustness testing across spatial resolution, statistical methodology, and ensemble comparison.
- **Legal:** The algorithm produces 137 majority-minority districts vs. 68 enacted (+69 surplus), with a 42% geographic concentration threshold identifying the feasibility boundary for VRA compliance.

## 2 Portfolio Architecture

The portfolio is organized into twenty active tracks. Track A synthesizes all other tracks; Tracks B–G generate the empirical, algorithmic, legal, and ensemble results that the original synthesis draws on; Tracks I–M extend the program to incumbency, apportionment, compactness, partisan fairness, and community character weights; Tracks N–S extend to population counting, democratic outcomes, reform pathways, 2030 planning, adversarial robustness, and statistical inference; Tracks T–U factor the overloaded algorithm material into plan construction and search/optimization.

| Track    | Theme                           | Key Headline Result                                                                                                                        | Papers |
|----------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <b>A</b> | Portfolio Synthesis             | 190+ papers synthesized across 20 active tracks; 50 states $\times$ 3 decades $\times$ +22% compactness at CV < 2%                         | 6      |
| <b>B</b> | Algorithm Foundations           | Recursive bisection, edge weighting, METIS formulation, complexity, n-way comparison, and seed-sensitivity foundations                     | 9      |
| <b>C</b> | Validation & Robustness         | Results robust across 130 $\times$ unit-count range; near-zero efficiency gap as byproduct; 30% more swing districts than enacted          | 10     |
| <b>D</b> | VRA & Legal Implementation      | +69 majority-minority district surplus; 42% geographic threshold; model federal statute formalizing Apportion-Regions                      | 9      |
| <b>E</b> | Experimental Extensions         | Geometric cost of federalism quantified; proportionality gain from multi-member variants; international applications                       | 8      |
| <b>F</b> | State Legislative Redistricting | Algorithm scales from 435 congressional to 400-district state chambers; NestSection achieves senate = 2 $\times$ house spine compatibility | 7      |
| <b>G</b> | Ensemble Methods & Comparison   | Bisection plan at 0.1–0.7th compactness percentile (WI/GA/PA/CA), 50th in NC; G.14 (practitioner ensemble comparison) accepted at 4.0/4    | 16     |
| <b>I</b> | Incumbency Effects              | Incumbent pairing probability, safe-seat dynamics, and legal treatment of incumbency as re-districting criterion                           | 5      |
| <b>J</b> | Apportionment Methods           | Huntington-Hill through Jefferson/D’Hondt comparison; bisect-apportion implementation verified against official 2020 apportionment         | 7      |

| Track    | Theme                       | Key Headline Result                                                                                                                                         | Papers |
|----------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <b>K</b> | Compactness Taxonomy        | Polsby-Popper, Reock, Convex Hull, Schwartzberg, Length-Width, population-weighted; multi-metric court guide                                                | 8      |
| <b>L</b> | Partisan Fairness Metrics   | Efficiency gap, mean-median, declination, seats-votes curve; proportionality paradox: $\sigma \approx 0$ for competitive states                             | 7      |
| <b>M</b> | Community Character Weights | Seven measurable signals (economic, land use, housing, commuting, topographic, administrative, transit) operationalizing communities of interest doctrine   | 9      |
| <b>N</b> | Population Counting         | Who counts in the balance metric: total pop vs. citizen VAP vs. prison-adjusted; N.5 definitive 50-state comparison; Evenwel v. Abbott implications         | 6      |
| <b>O</b> | Outcomes & Representation   | Does compact redistricting improve democracy? RQI: bisect plans 0.62 vs. 0.41 enacted; +23pp competitive districts; $-0.07$ DW-NOMINATE; $-14$ min distance | 6      |
| <b>P</b> | Reform Pathways             | Federal legislation (DIA), ballot initiatives (CA/CO/MI), commission integration, court-ordered algorithmic remedies, implementation costs ( $< \$50K$ )    | 6      |
| <b>Q</b> | 2030 Forward Analysis       | Demographic projections (TX+3, NY-2), data infrastructure (43 states need BG), TX $k = 41$ prime fallback proved, ratio-optimal recommended                 | 5      |

| Track    | Theme                  | Key Headline Result                                                                                                                            | Papers |
|----------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <b>R</b> | Adversarial Robustness | Five gaming vectors; combined max < 0.5 seats nationally; all detectable via SHA-256 audit chain; certificate pinning limitation documented    | 5      |
| <b>S</b> | Statistical Inference  | ESS-corrected permutation test (p=0.004 for NC at 10K steps); Bayesian Detection Score via Beta posterior; FDR control on 1,050 L-track tests  | 5      |
| <b>T</b> | Plan Construction      | GeoSection, AreaSection, ApportionRegions, VRA-Section, CVD, BFS Growth, and Moving-Knife structure methods factored out of the former B track | 13     |
| <b>U</b> | Search & Optimization  | ConvergenceSweep, simulated annealing, tempering, ILP certificates, Pareto/NSGA-II, and former Track H search strategies                       | 11     |

Table 1: Portfolio track summary (190+ papers across twenty active tracks A–G, I–U).

### Three-Layer Compositor

Every redistricting run is governed by three orthogonal choices that correspond to the three-layer compositor of the `bisect` binary:

| Flag                            | Controls                 | Key Variants                                 |
|---------------------------------|--------------------------|----------------------------------------------|
| <code>--structure</code>        | Bisection tree shape     | standard-bisect, prime-factor (ApportionR)   |
| <code>--weights-override</code> | METIS edge/vertex signal | geographic (default), county, unweighted, vr |
| <code>--search</code>           | Seed-selection strategy  | single, convergence (T=600), percentile, b   |

Track B addresses algorithm foundations; Track T addresses plan-construction and structure variants; Track G addresses ensemble algorithms; Track U addresses search, optimization, certification, and algorithm selection.

## 3 Reading Paths

The following curated reading paths guide each audience to the papers most relevant to their interests. Papers are cited by track code (e.g., B.1, C.5). Paper track codes correspond directly to the folder names under `research/`.

### 3.1 For Political Scientists

1. **B.1** (Recursive Bisection): Establishes the impossibility defense—the algorithm cannot see partisan data—and demonstrates +22% compactness improvement over enacted maps across 50 states and three census decades.
2. **D.0** (VRA Compliance): Documents the 137 vs. 68 majority-minority district count, establishing that algorithmic redistricting exceeds current VRA performance by 69 districts without explicit racial targeting.
3. **D.1** (Threshold Analysis): Identifies the 42% minority concentration threshold as the geographic limit of algorithmic VRA compliance, with direct implications for the Gingles preconditions.
4. **C.5** (Efficiency Gap Analysis): Shows that near-zero partisan efficiency gap emerges as a *byproduct* of compactness optimization across all 50 states, without partisan input.
5. **C.8** (Competitive Elections): Demonstrates that algorithmic maps produce 30% more swing districts than enacted plans, addressing the competitive-elections objection directly.
6. **T.4** (ApportionRegions): Presents the prime-factor bisection tree producing NC 7D/7R and 223D/209R nationally—the clearest demonstration that geographic structure drives partisan outcome.
7. **A.0** (Synthesis Metapaper): Integrating account of all 60+ empirical results, written for a *Science*-level general audience.

### 3.2 For Computer Scientists

1. **B.2** (Edge-Weighted Bisection): Introduces geographic proximity weights lifting compactness an additional +22% beyond B.1’s baseline, explaining why single-objective edge-cut dominates.
2. **B.3** (Multi vs. Edge): Proves that single-objective edge-cut dominates multi-constraint METIS for redistricting—the central algorithm-design finding of the program.
3. **B.6** (Computational Complexity): Proves NP-hardness of balanced planar bisection and characterizes METIS runtime as  $O(n_{\text{iter}} \times n \log k)$  with empirical near-linear scaling ( $O(n^{1.07})$ ).
4. **B.7** (Solution Space): Establishes solution-space tightness: seed CV < 2% for 96% of states, partisan variance near zero across all runs.
5. **G.14** (Ensemble Comparison): Practitioner comparison of redistricting ensemble algorithms; highest-scored paper in the portfolio at 4.0/4.
6. **U.10** (Redist-Ensemble): Presents the Rust ReCom implementation achieving 2500× speed over the Python GerryChain baseline.
7. **T.10** (Centroidal Voronoi): Geometric district construction via CVD; accepted at 3.9/4. T.11 extends to Euclidean distance on EPSG:5070 projections for 2–6% additional Polsby-Popper gain in coastal and elongated states.

### 3.3 For Legal Scholars

1. **B.1** (Recursive Bisection): The impossibility defense and its implications under *Rucho v. Common Cause*; Huntington-Hill as precedent for mathematical governance.
2. **B.02** (One Federal Law): Formalizes ApportionRegions as the one-sentence geographic completion of Huntington-Hill; the model federal statute.
3. **D.0** (VRA Compliance): Full VRA Section 2 compliance analysis: 137 vs. 68 majority-minority districts and the totality-of- circumstances framework.
4. **D.1** (Threshold Analysis): The 42% geographic threshold and its relationship to the Gingles compactness precondition.
5. **D.3** (Compactness Tradeoff): VRA–compactness tradeoff characterized: real but bounded, with most VRA districts achievable within 85% of baseline compactness.
6. **D.4** (Legal Implementation): 50-state adoption pathway analysis; the model statute that formalizes ApportionRegions as the statutory vehicle.
7. **D.5** (Gingles Bloc-Voting): Expert witness methodology for the Gingles bloc-voting analysis required when algorithmic plans face Section 2 challenge.

### 3.4 For Practitioners (Redistricting Commissions)

1. **B.1** (Recursive Bisection): Core method overview; compactness and partisan neutrality at national scale.
2. **B.2** (Edge-Weighted Bisection): Implementation details for compact districts using geographic proximity weights.
3. **T.4** (ApportionRegions): Prime-factor bisection tree for balanced partisan outcomes; the algorithm most directly usable by commissions.
4. **U.1** (Convergence Sweep): The statutory seed formula ( $T=600$ ) balancing computational cost against partisan variance reduction.
5. **U.8** (PercentileSweep): Statutory legal posture selection via percentile position in the ensemble—the practitioner-facing search strategy.
6. **C.9** (Adoption Case Studies): Arizona, California, and North Carolina adoption pathway analysis; political feasibility documentation.
7. **A.5** (Policy Brief): 4-page brief targeting legislative staff and redistricting commissioners; entry point for non-technical audiences.

### 3.5 For Judges and Legislative Staff

1. **A.2** (Portfolio Summary): Rapid briefing document condensing the headline findings across all 8 tracks.
2. **A.5** (Policy Brief): 4-page accessible summary for non-technical decision-makers.
3. **B.02** (One Federal Law): The one-sentence geographic completion of Huntington-Hill and its constitutional grounding.
4. **D.4** (Legal Implementation): Model federal statute and 50-state adoption pathways.
5. **D.5** (Gingles Bloc-Voting): The evidentiary methodology courts require when algorithmic plans face VRA challenge.

6. **C.6 (User Study):** Evidence that algorithmic maps are rated fairer by the public; legitimacy basis for adoption.

## 4 Glossary

### **ApportionRegions**

Prime-factor bisection tree structure that splits the state into sub-regions corresponding to the prime factors of the district count, enabling balanced geographic and partisan outcomes. Formalized as a one-sentence federal statute in B.02.

### **Bisection tree**

The hierarchical sequence of binary splits that defines how a state is partitioned into districts under recursive bisection. Structure variants (standard, prime-factor, ratio-optimal) are the subject of Track T plan-construction sub-track.

### **Census tracts**

Geographic units used as the atomic partitioning units ( $\sim 4,000$  residents each); the building blocks from which districts are assembled.

### **Compactness**

A geometric measure of district shape regularity. Polsby-Popper ( $4\pi \times \text{Area} / \text{Perimeter}^2$ , range  $[0, 1]$ ) is the primary measure used throughout the portfolio; higher values indicate more compact districts.

### **Edge-weight**

A numerical cost assigned to the graph edge connecting two adjacent census tracts. Higher weights make an edge expensive to cut, encouraging the partitioner to preserve adjacency between similar tracts, which produces more compact districts.

### **Efficiency gap**

Partisan fairness measure: the difference in wasted votes between parties divided by total votes. A near-zero efficiency gap emerges as a byproduct of compactness optimization (C.5).

**Ensemble** The distribution of all valid redistricting plans consistent with statutory constraints, sampled by MCMC methods (GerryChain, ReCom, forest-ReCom, etc.). Track G positions the bisection plan within this distribution.

### **GerryChain / ReCom**

Open-source Python toolkit (Metric Geometry and Gerrymandering Group) for MCMC redistricting ensemble sampling. GerryChain implements ReCom (recombination) proposals. Track G compares bisection plans to GerryChain-generated ensembles.

### **Gingles test**

Three-part legal test from *Thornburg v. Gingles* (1986) establishing preconditions for a successful VRA Section 2 claim: (1) minority group sufficiently large and geographically compact; (2) minority group politically cohesive; (3) majority votes as a bloc to defeat minority candidates.

### **Impossibility defense**

The core philosophical claim of the portfolio: the METIS recursive bisection algorithm cannot be instructed to gerrymander by construction because it is not given access to partisan data. It minimizes edge cuts in a geographic graph—a purely geo-



metric objective. Compact neutral maps can still reflect geographic voter concentration, as shown in Track C.5.

**Majority-minority district**

A district in which minority voters constitute at least 50% of the voting-age population (VAP), satisfying the Gingles compactness precondition for VRA Section 2 protection.

**METIS** Multilevel  $k$ -way graph partitioning library (Karypis & Kumar, 1998). The core computational engine used throughout the portfolio to bisect the census tract adjacency graph.

**NestSection**

Bisection tree structure enforcing senate =  $2 \times$  house spine compatibility for bicameral redistricting (T.6, F.2).

**PercentileSweep**

Search strategy (U.8) selecting among valid plans by their partisan percentile position in the ensemble, enabling commissions to choose a legal posture without targeting a specific partisan outcome.

**Polsby-Popper**

See *Compactness*.

**VRASection**

Bisection structure variant (T.7) that aligns bisection cuts with minority geographic concentration, enabling VRA compliance post-*Allen v. Milligan* (2023) and *Callais* (2025).

## 5 Code and Data

- **Primary tool:** The `bisect` Rust binary, built from the `crates/` workspace in this repository. It is the production-grade implementation of all three-layer compositor options described in Section ?? and runs approximately  $213 \times$  faster than the archived Python pipeline.
- **Replication package:** See A.4 for the AEA-compliant replication materials package, including configuration files, census data download scripts, and output reproducibility documentation.
- **Census data:** U.S. Census Bureau redistricting files for 2000, 2010, and 2020, downloaded via the `bisect fetch` command. Raw data is not committed to the repository ( $\sim 55$  GB).
- **METIS library:** <http://glaros.dtc.umn.edu/gkhome/metis/metis/overview>. The portfolio also includes a pure-Rust METIS implementation (`bisect-metis` crate) for portable binary distribution.
- **Contact:** [giodl@microsoft.com](mailto:giodl@microsoft.com)